

BRUNO HENRIQUE FREITAS DE MELO

SOFTWARE ENGINEER · BACKEND & FULLSTACK

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SUMMARY

Software Engineer specializing in backend architecture and fullstack development, with extensive expertise in RESTful APIs (Spring Boot, Django REST Framework), relational modeling, and concurrent data processing. Experienced in building production systems with contextual RBAC via Spring Security, high-throughput concurrent pipelines using ThreadPoolExecutor (136k+ records collected with zero failure rate), and 30+ endpoints documented via OpenAPI/Swagger. Published researcher at BrasNAM (SBC, 2025) on social media text mining. M.Sc. candidate in Computer Science (UFAC) pursuing a Postgraduate Specialization in AI-Driven Test Engineering (UFAC/Motorola).

PROFESSIONAL EXPERIENCE

Web Academy — Motorola / UFAC Partnership

June 2025 – December 2025

Fullstack Developer & Scrum Master (Hands-On)

- Modeled a relational domain in MySQL with 6 interdependent entities (users, projects, partnerships, activities, tasks, and delivery deliverables), writing custom JPA queries with pagination and dynamic filtering.
- Implemented JWT authentication and project-level contextual authorization via Spring Security, establishing dynamic RBAC where user privileges adapt according to the accessed scope.
- Documented and standardized 30+ REST endpoints via OpenAPI/Swagger, establishing consistent API response contracts and error-handling standards.
- Diagnosed and resolved a critical multipart asynchronous file upload incident caused by HTTP parameter inconsistencies, achieving zero downtime.
- Served as Scrum Master: facilitated backlog grooming, coordinated sprints, and aligned technical deliverables with stakeholders.

National Institute of Social Security (INSS)

August 2024 – November 2024

Backend Developer Intern

- Developed a RESTful API with Django REST Framework for asset management and maintenance auditing across 9 business entities (equipment, branch offices, civil servants, and work orders).
- Operated with complete autonomy from initial requirements gathering with public officials to the delivery and sign-off of the approved prototype.
- Implemented JWT authentication and standardized 100% of API response contracts and error-handling routines.
- Built bulk CSV export routines for seamless integration with internal administrative spreadsheets.

PROJECTS

Concurrent Bluesky Mining Pipeline — 2024 Wildfires

2024 – 2025

Researcher | Paper published at BrasNAM/SBC (sol.sbc.org.br)

- Architected a concurrent data collection pipeline using Python and ThreadPoolExecutor, mining over 136k posts with a zero failure rate, featuring automated JWT token refresh and exponential backoff retries.
- Applied semantic preprocessing with spaCy/NLTK and topic modeling via LDA (Gensim), resulting in a peer-reviewed paper published by the Brazilian Computer Society (SBC).

RefAgent-Py — Automated Code Refactoring with Multi-Agent Systems in Python

2026

Researcher | PPGCC/UFAC | github.com/I-Am-BrunoHFMelo/RefAgent-Python

- Ported the four-agent architecture of the RefAgent framework (Planner, Generator, Compiler, and Tester) from Java to Python, replacing static compilation (javac) with decoupled dynamic quality gates (py_compile, mypy, and Pylint with relative baselines), orchestrated via Ollama with local open-weight models.
- Multi-agent orchestration boosted functional test pass rates by up to 62% over single-agent baselines on Flask production files, benchmarked across 16 RefactorBench tasks and 17 real-world files on local GPU infrastructure (RTX 5080, PPGCC/UFAC).

Deck & Dice — Native Android App for Tabletop RPGs

2026

Android Developer | github.com/I-Am-BrunoHFMelo/Deck-Dice

- Developed a native Android app in Java (SDK 34) featuring a custom dice expression parsing engine (XdY, khN/klN, modifiers, and success checks), 2D physics-based rendering with custom Canvas, and persistence via Room Database.
- Implemented a Card Wallet for custom roll combos, persistent history of the last 100 rolls, session notes, and dynamic themes (Medieval, Cyberpunk, Horror, Forest).

Anima Ultima — Offline-First Companion App for Fabula Ultima TTRPG

2026

Mobile Developer | github.com/I-Am-BrunoHFMelo/Anima-Ultima-Aplicacao-Mobile-para-Fabula-Ultima

- Engineered a responsive, offline-first companion app using Ionic 8 + Angular 20 + Capacitor, featuring autosave via Dexie.js (IndexedDB) with a 600ms debounce and Retro Pixel Art UI (NES.css).

- Implemented complex RPG game mechanics: character sheets with automated HP/MP/IP calculation, rank-based bestiary with elemental affinities, session tracker with XP and conflict clocks, and a magic ritual calculator.

Fuzzy Inference System for Social Post Sentiment & Engagement Analysis

2026

Author | github.com/I-Am-BrunoHFMelo/fuzzy-tweet-temperature

- Implemented a 27-rule Mamdani fuzzy system correlating polarity (VADER), subjectivity (TextBlob), and engagement with logarithmic compression, eliminating rigid threshold boundary artifacts from standard if-else logic.

Event-Driven Mini-CRM with Laravel Reverb

2026

Backend Developer | github.com/I-Am-BrunoHFMelo/laravel-mini-crm-contatos

- Built a CRM API featuring queued asynchronous jobs, model observers for automated notifications, and real-time WebSocket broadcasting with Laravel Reverb.

EDUCATION

M.Sc. in Computer Science

2026 – 2028 (In Progress)

Federal University of Acre (UFAC)

IARTES — Postgraduate Specialization in AI-Driven Software Testing

2026 – 2027 (In Progress)

Federal University of Acre in partnership with Motorola

B.S. in Information Systems

2019 – 2025 (Completed)

Federal University of Acre (UFAC)

Web Academy — Full-Stack Web Development Training

2025 (Completed)

Federal University of Acre in partnership with Motorola

N.A.V.E. Tech Acre — Advanced Hub for Entrepreneurship and Technology

2022 (Completed)

Federal University of Acre in partnership with Samsung

TECHNICAL SKILLS

Backend & APIs: Java (Spring Boot, Spring Security, JPA/Hibernate, Maven), Python (Django REST Framework, Django ORM), JWT, Project-Contextual RBAC, OpenAPI/Swagger, SOLID Principles.

Data, AI & Research: Concurrent Processing (ThreadPoolExecutor, exponential backoff, rate limit handling), NLP (spaCy, NLTK, VADER/LeIA, Gensim, LDA), Fuzzy Inference Systems (Mamdani, scikit-fuzzy), Multi-Agent Systems with LLMs (Python, Ollama, local open-weight models).

Frontend & Mobile: Angular 19 + TypeScript (Standalone Components, Route Guards, Interceptors, RxJS), Ionic 8 + Capacitor (Android/Web), Native Android — Java (Room Database, custom 2D Canvas, Material Design), HTML5, CSS3, Tailwind CSS.

Databases: MySQL, PostgreSQL, SQLite, Room Database (Android), IndexedDB (Dexie.js), JPA/Hibernate, Django ORM, custom queries, dynamic pagination.

DevOps & Tools: Docker, CI/CD Pipelines, GitHub Actions, Git/GitHub (branching, PRs, code review), Ollama (local LLM inference), IoT — Internet of Things (N.A.V.E. Tech, 2022).

Testing & Software Quality: TDD (Test-Driven Development), JUnit (Java), PyTest / PyUnit (Python), mypy, Pylint, Software Quality, AI-Driven Test Engineering.

Methodologies & Processes: Scrum, Requirements Engineering, Design Thinking, Design Sprint, Lean Startup, Requirements Gathering, Scientific Research.

LANGUAGES

Portuguese — Native.

English — Intermediate (B1): technical documentation reading, professional writing, and written communication.